A.

I propose the narrative from 'ATBS Implementation' to 'Designing a Domain-Specific Architecture direction.
By focusing on how our PS/PL partitioning and custom on-chip buffers eliminate off-chip DDR access.
we elevate this from an application paper to a core computer architecture contribution to solving the System-Level Memory Wall and Latency Wall Issues

(ResNet50, MobileNet) = CNNs
(ATBS+FRM) = Vision Algo

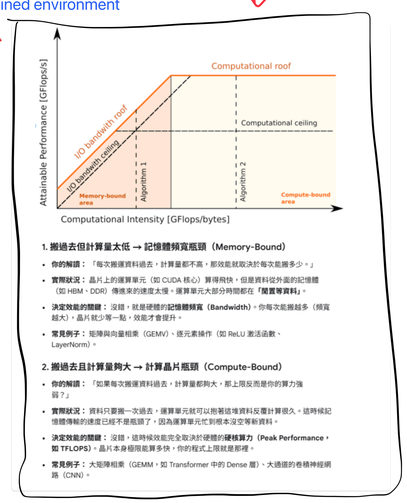
(X) image → store in DDR memory → FTM+ATBS on x86CPU or ARM CPU for ROI → store in DDR memory → NPU, GPU take data from outter DDR memory for CNNs

(O) image → BRAM as LineBuffer → FTM+ATBS on FPGA-PL → BRAM as Ping_Pong Buffer (get ROI, feed ROI) → FPGA PS/PL partition for CNNs with energy efficient method design

B.

- Title: An Energy-Efficient Domain-Specific Architecture for Hybrid Vision-CNN Pipelines on FPGA
- Observation
  - Observation(s): hybrid datapath (FDM+ATBS + CNN) is highly accurate for edge object detection.
  - Issue: pure NPUs (like Hailo-8) cannot run FDM well . Relying on Host CPU causes heavy off-chip DDR data transfers, lead to Memory Wall , Latency Wall.
- Objective
  - Objective(s): design an Domain-Specific Architecture on FPGA with a custom hardware pipeline to achieve zero-Copy, eliminating DDR memory bottlenecks.
  - Model: ResNet50 (Compute-bound) vs. MobileNet (Memory-bound) as two extreme hardware stress tests in FDM+ATBS state.
- Challenge: design custom BRAM buffers and PS/PL partitioning to prevent AXI bus bottlenecks.
- Method
  - Main Idea & Insight: the movement of data costs more power than computation. fusing FDM and part of CNNs into a single PL pipeline saves immense energy.
  - Method(s): 3 Hardware platforms (FPGAsoc, GPU + ARM CPU, x86 CPU + ASIC) × 2 Models (ResNet, MobileNet).
  - Justify: elevates the work from "algorithm implementation" to "computer architecture design level" to discuss adaptability in Compute-bound and Memory-bound extreme environment with different hardware plateforms.
- Result
  - Analyze: FPS(frame per sec)/Watt (Energy Efficiency), Time-to-First-Inference/output results (Latency) , and roofline model
  - Evaluate: prove FPGA beats ASIC/GPU in system level energy efficiency on ResNet and MobileNet models.
  - Conclusion: conclude a hardwares and models deployment guide in a resource constrained environment
- Reference
  - List: x2
  - Paper: DSA design & pratik for PL application, Stanford.. a New Golden Age for CA

roofline model



Low Data Reuse (∵ DSCC)
↓
Low Arithmetic Intensity
↓
Memory Bound (bandwidth)
↓
face for Memory Wall
due to frequent DDR transition

High Data Reuse (∵ FC)
↓
High Arithmetic Intensity
↓
Compute Bound Type
↓
face for Latency Wall
if combining with ASIC